SMARTX: Porewater Methane

May 2025 Samples

2025-10-30

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##Run Information

```
cat("Run Information: Brenden D. Blakley ") #lets you know what section you're in
```

```
## Run Information: Brenden D. Blakley
```

```
#set the run date & user name
sample_year <- "2025"
sample_month <- "May"
user <- "Brenden Blakley"
#identify the files you want to read in
#read in as a list to accommodate multiple runs in a month

GHG_files <- c("Raw Data/20250624_GHGdata_SMARTX.csv","Raw Data/20250813_GHGdata_SMARTX.csv","Raw Data/2


# Define the file path for QAQC log file - NO Need to change just check year
log_path <- "Processed Data/SMARTX_ShimadzuGC_QAQC_2025.csv"


# Define final path for data be sure to change year and month and run number
final_path <- "Processed Data/GCReW_SMARTX_Porewater_CH4_202505.csv"

#Gas in Water Calculations
# "TempC" column value in Celsius (equilibration temperature) and volume of gas (in liters) in syringe
TempC = 25 #Equilibration Temperature (Celsius)

volume_g = 0.015 #volume of gas (in liters) in syringe when equilibrized
volume_w = 0.015 #volume of water (in liters) in syringe when equilibrized (should be equal to volume_g

#record any notes about the run or anything other info here:
run_notes <- ""

#Collection_Dates = "Raw Data/Sample_Collection_Dates_2024.csv"

cat(run_notes)
```

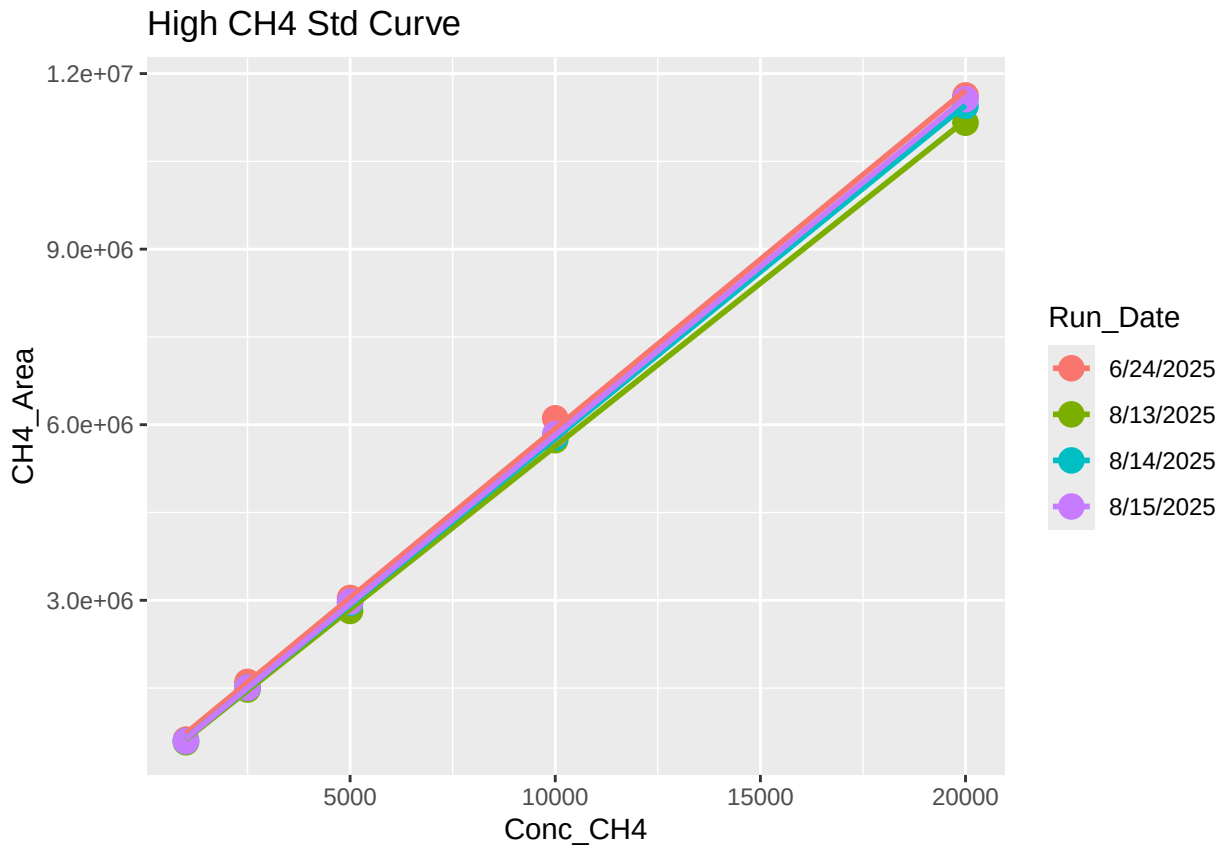
##Set Up

##Read in metadata and create similar sample IDs for matching to samples

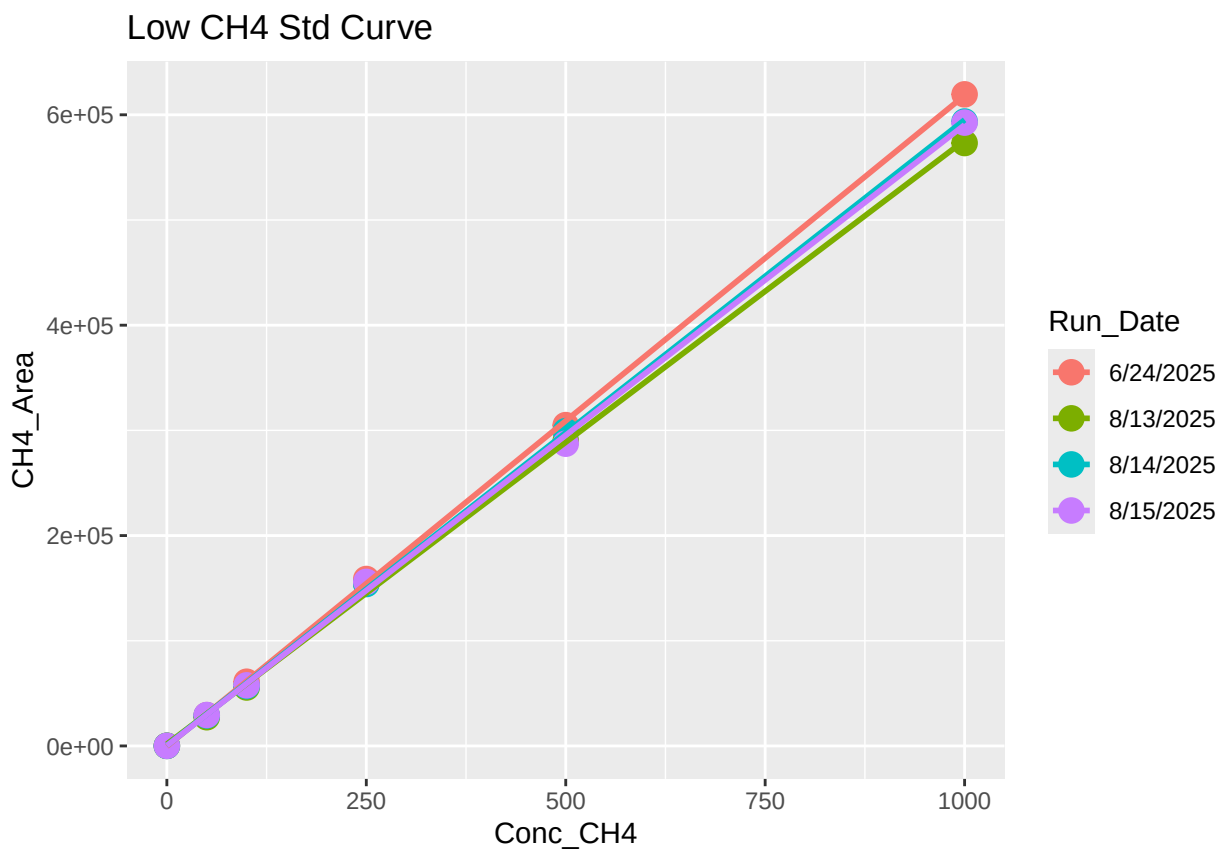
1 Read in data files

2 Assess standard curves

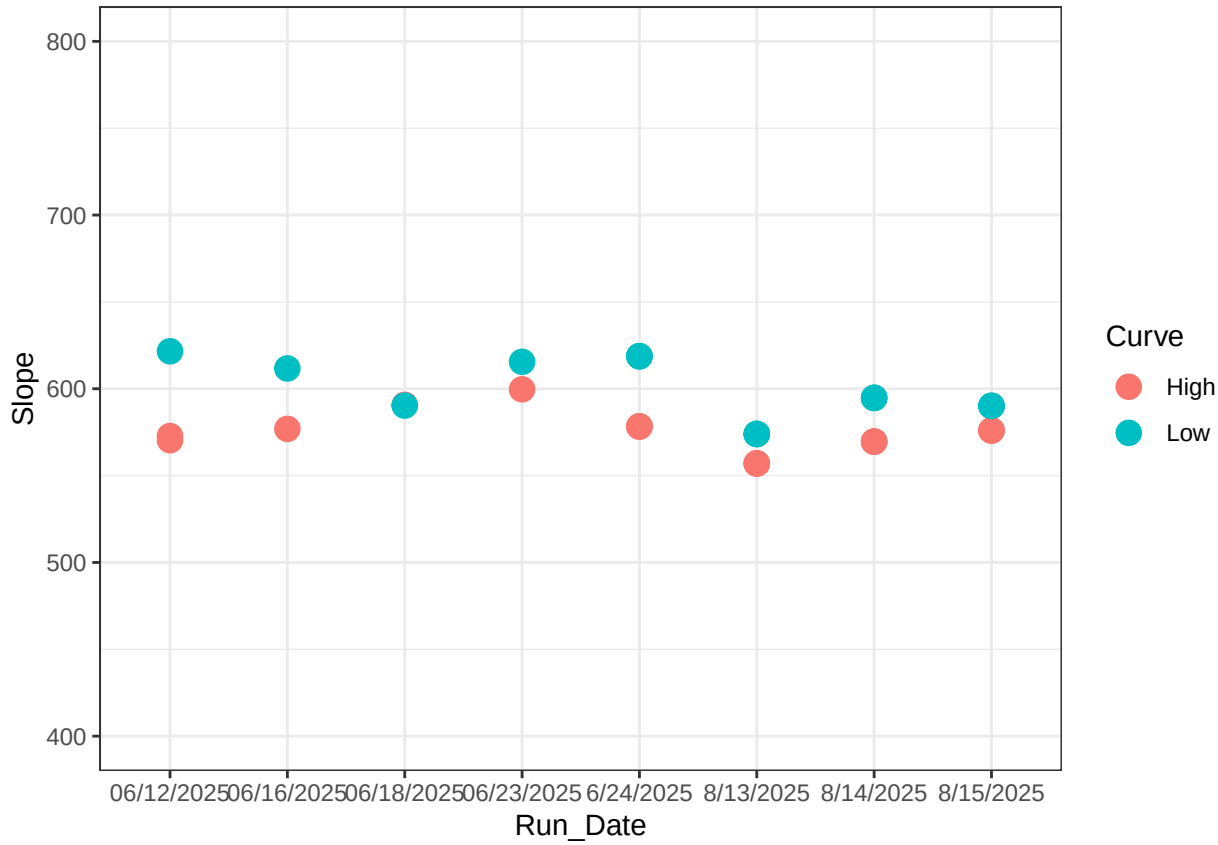
```
## `geom_smooth()` using formula = 'y ~ x'
```




```
## `geom_smooth()` using formula = 'y ~ x'
```




```
##   X Curve        R2    Slope    Intercept   Run_Date
## 1 1  High 0.9986295 576.8990  173986.2174 06/16/2025
## 2 2   Low 0.9990260 611.6961   -3791.2684 06/16/2025
## 3 3  High 0.9999101 570.3925  138747.3478 06/12/2025
## 4 4   Low 0.9979098 621.5734    3224.2535 06/12/2025
## 5 5  High 0.9989882 590.7374 -109278.7391 06/18/2025
## 6 6   Low 0.9990995 590.3138     954.7682 06/18/2025
```



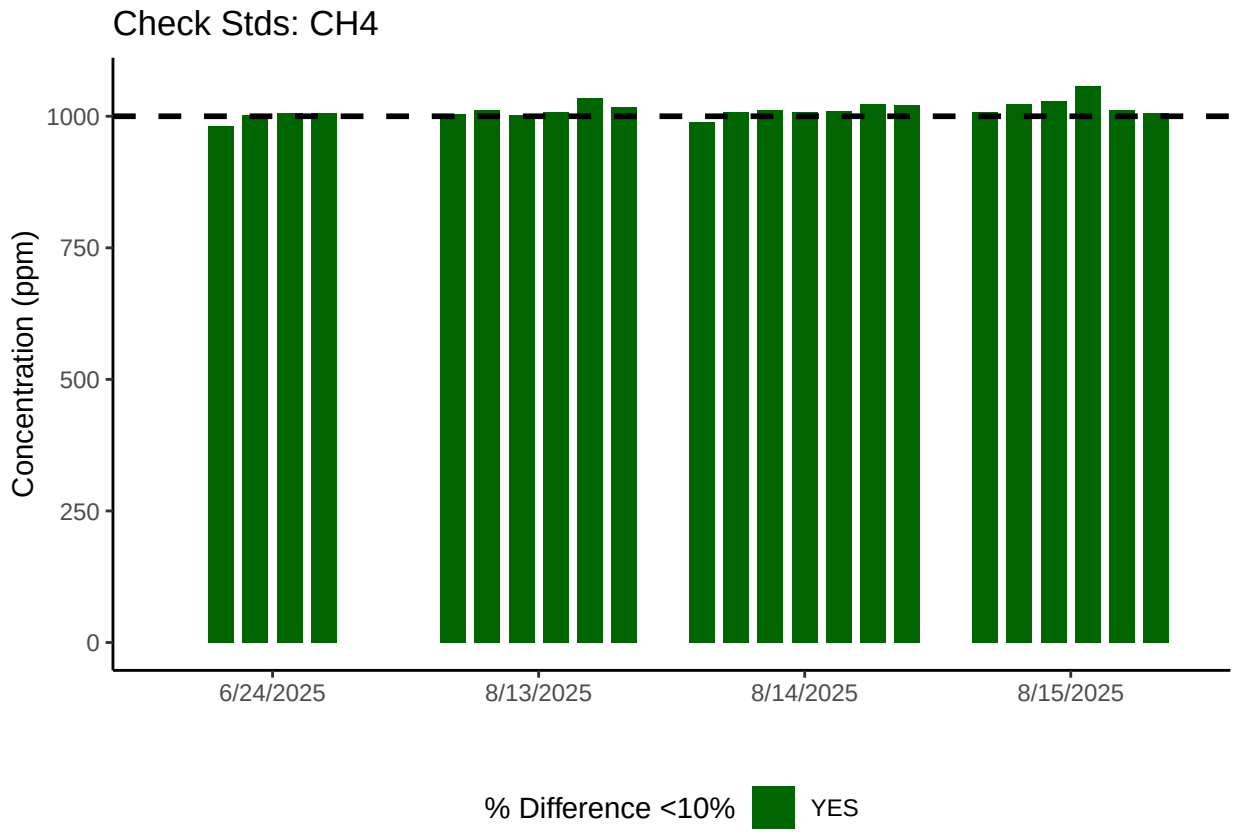

```
## [1] "High_R2 is Good-6/24/2025" "High_R2 is Good-8/13/2025"
## [3] "High_R2 is Good-8/14/2025" "High_R2 is Good-8/15/2025"
```

```
## [1] "Low_R2 is Good-6/24/2025" "Low_R2 is Good-8/13/2025"
## [3] "Low_R2 is Good-8/14/2025" "Low_R2 is Good-8/15/2025"
```

2.1 Now calculate the CH4 in ppm

3 Aanlayze the Check Standards

```
## >60% of CH4 Check Standards are within range of expected concentration - PROCEED
```



4 Analyze the Blanks

```
## Assess Blanks
```

```
## >60% of CH4 Blanks are below the lower 25% quartile of samples - PROCEED
```

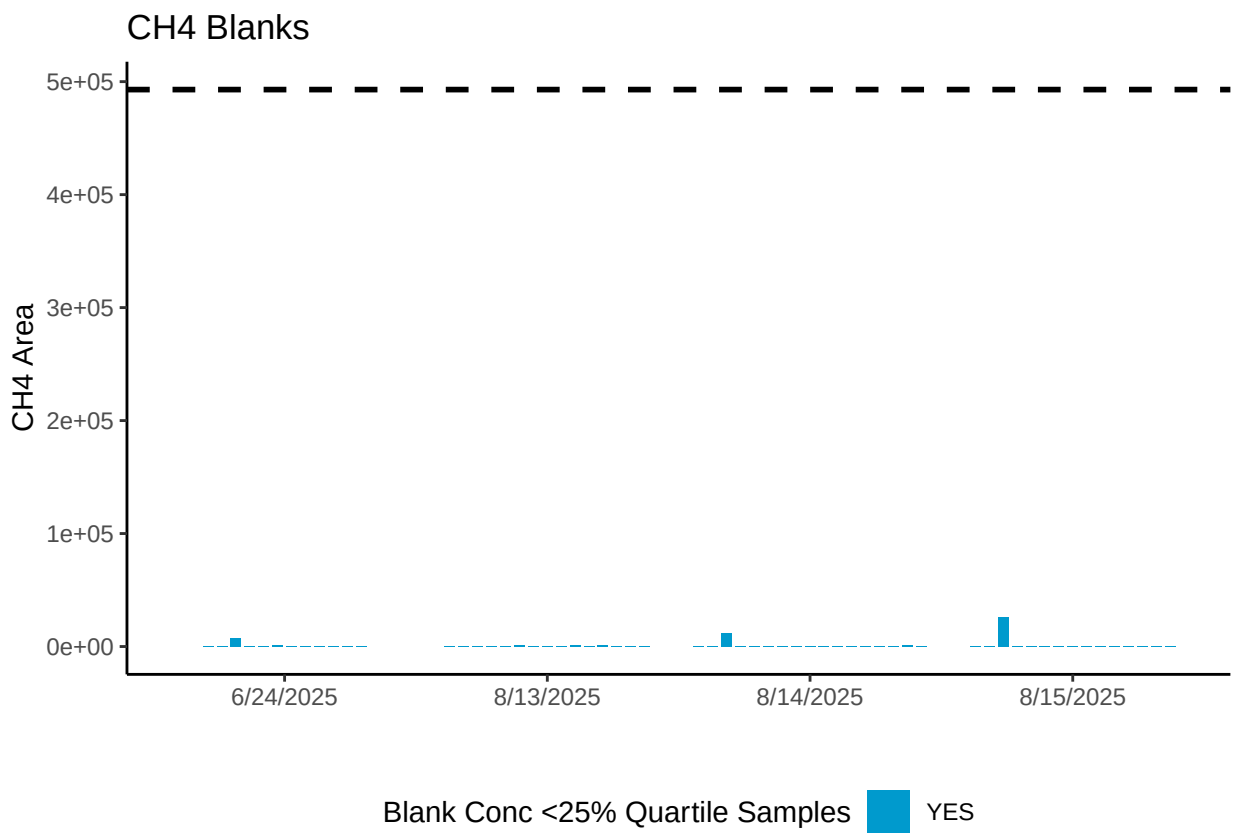
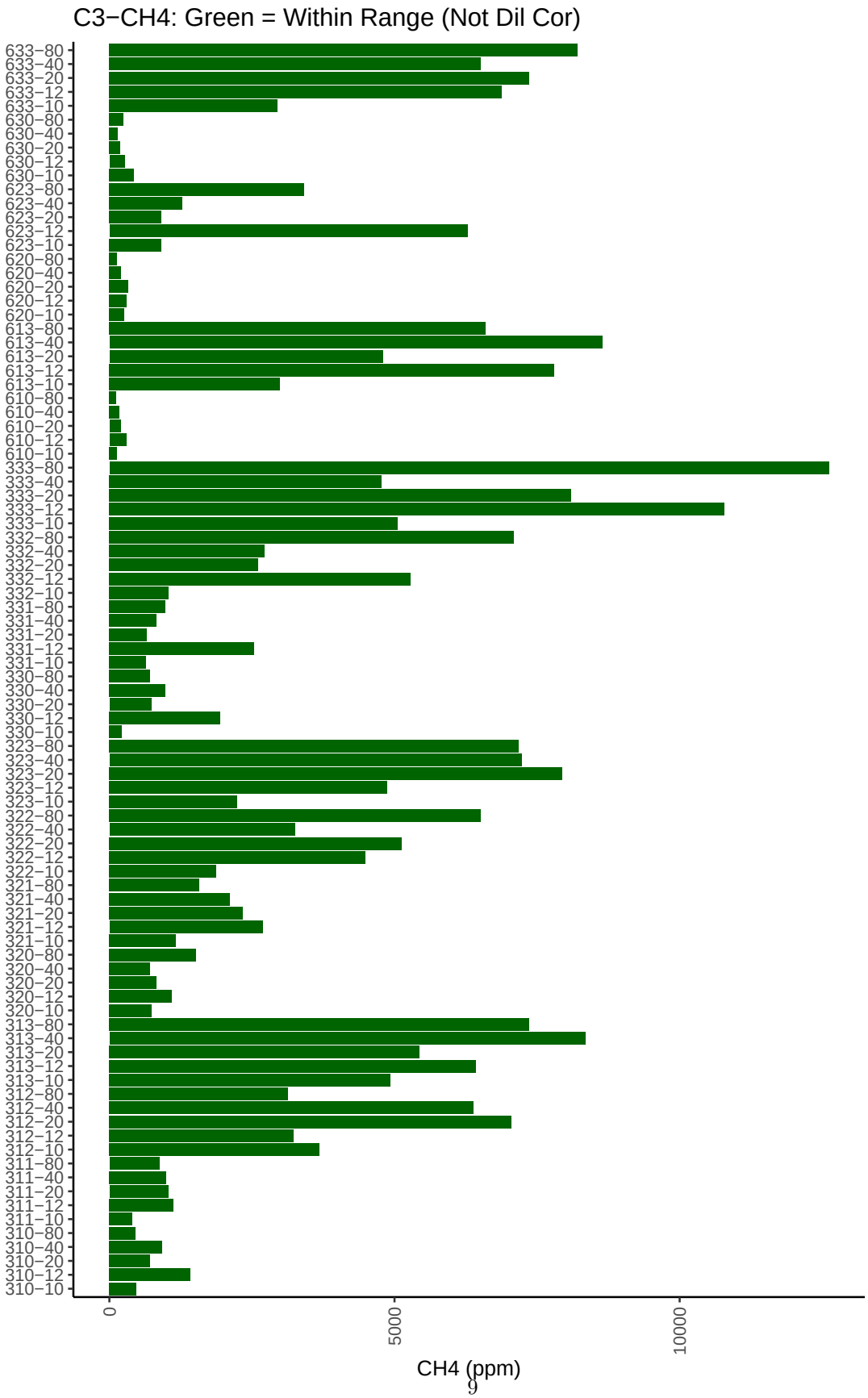
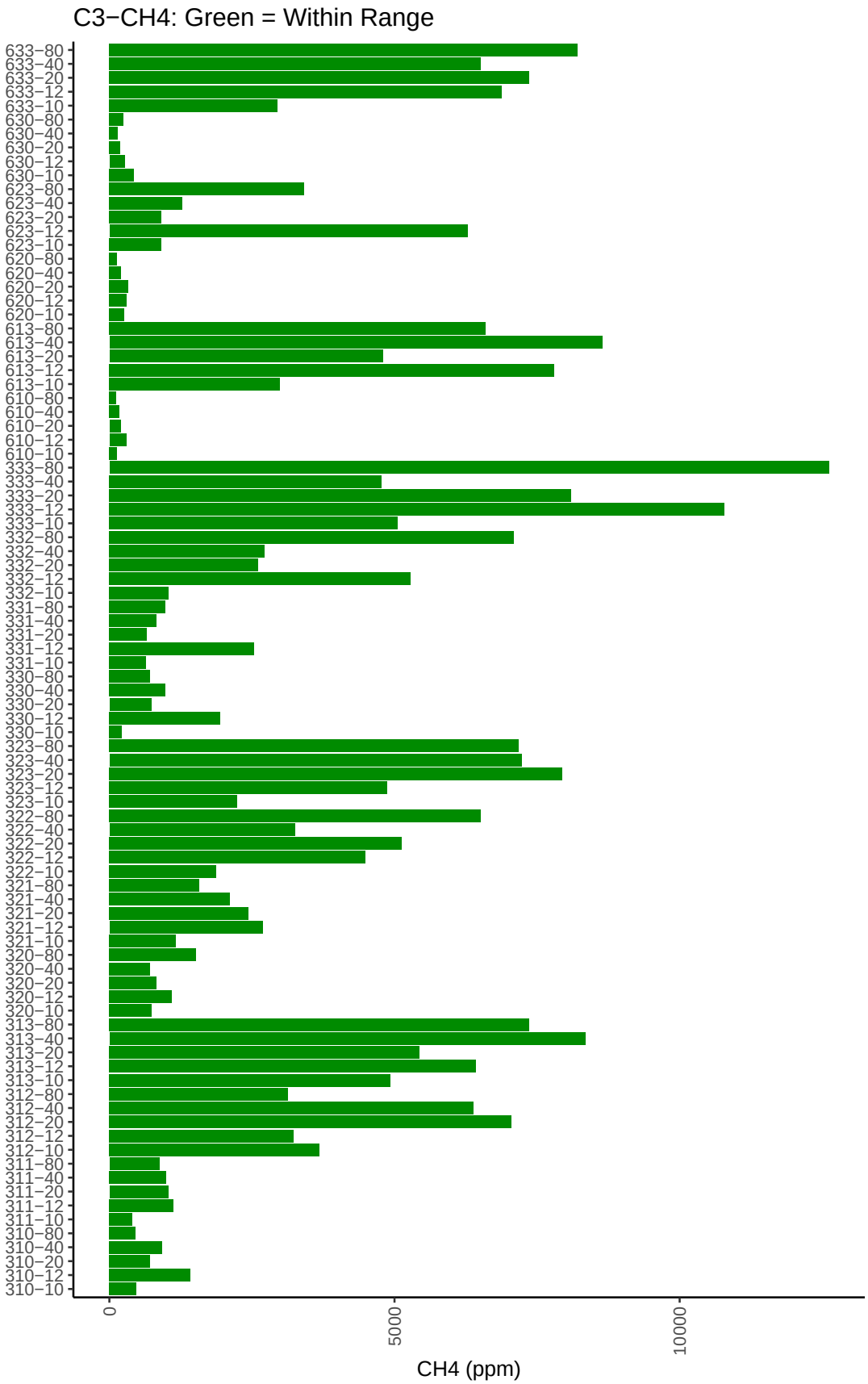


Table 1: Mean Area of Blanks

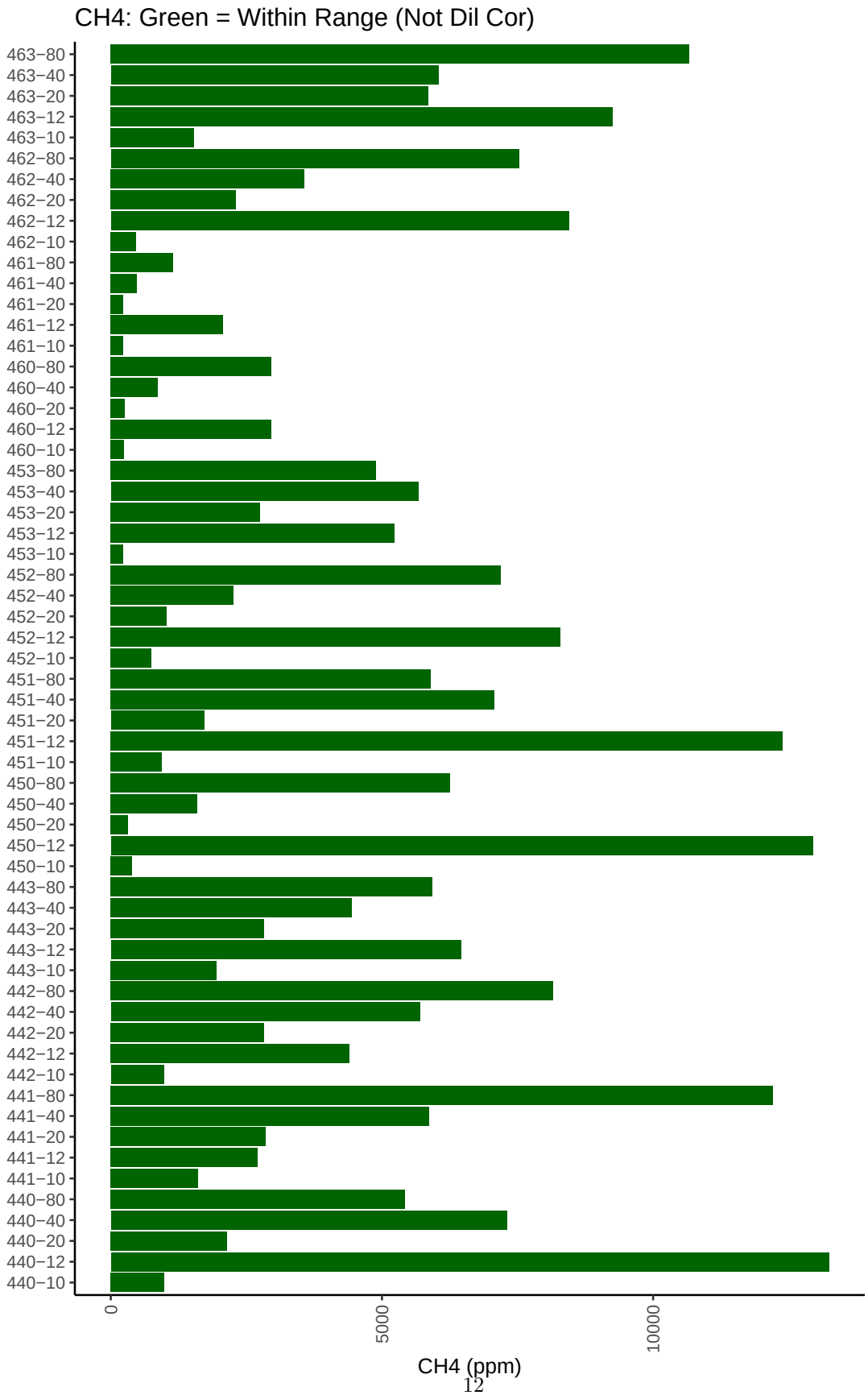
Test	Blank_Mean_Area
CH4	1176.763

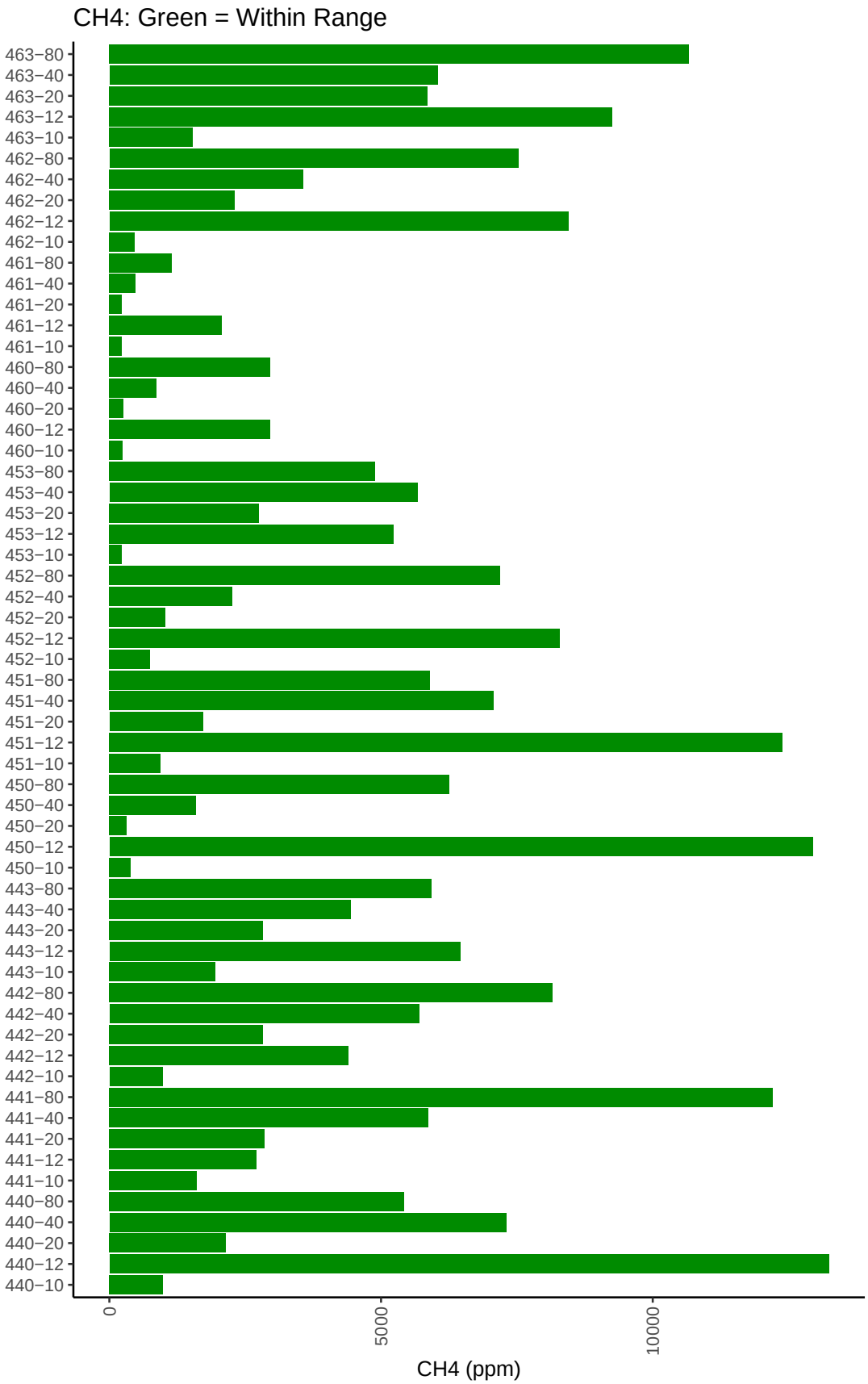
5 C3: Dilution correct samples and Check Range



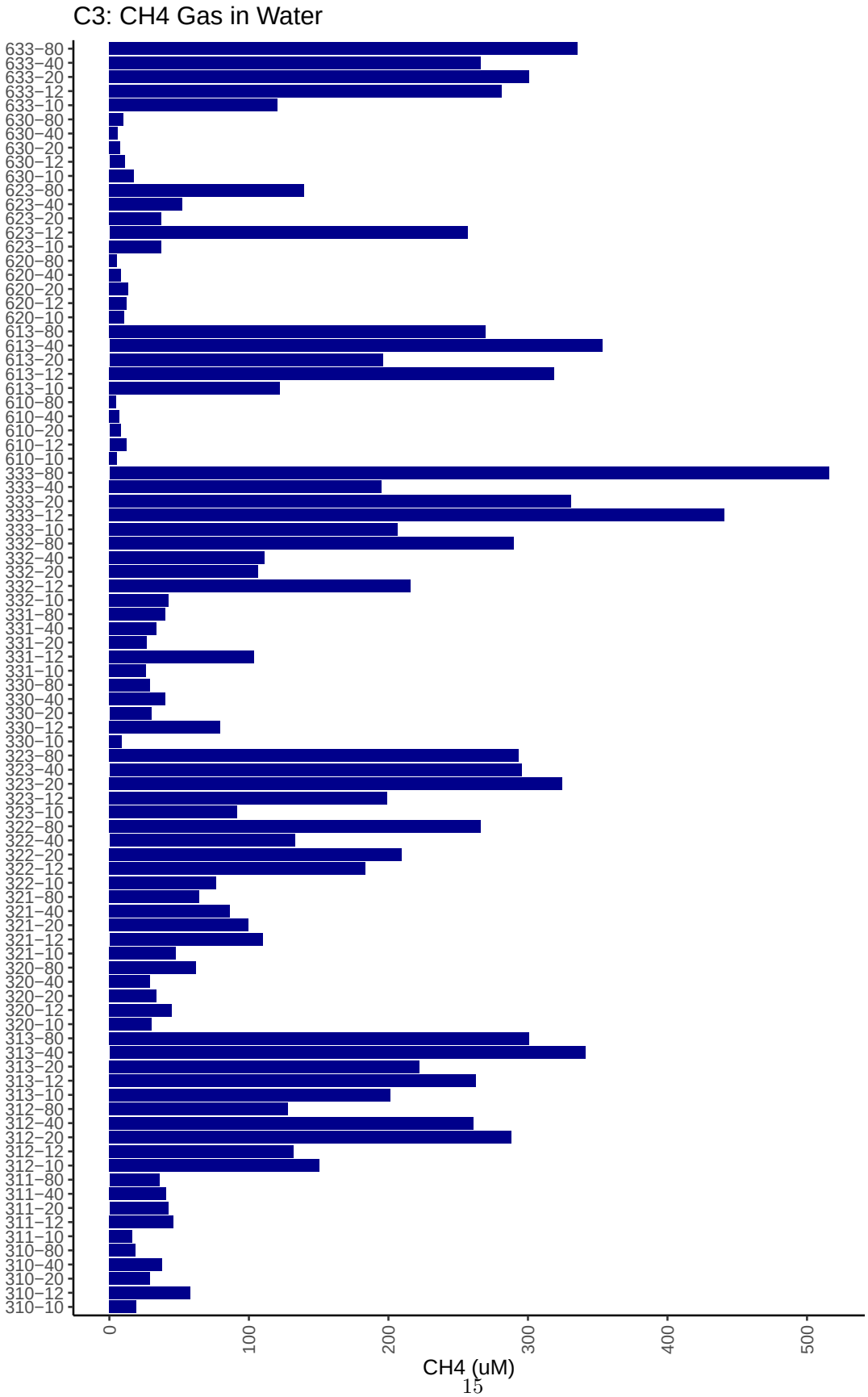

C3−CH4: Green = Within Range
633−80
633−40
633−20
633−12
633−10
630−80
630−40
630−20
630−12
630−10
623−80
623−40
623−20
623−12
623−10
620−80
620−40
620−20
620−12
620−10
613−80
613−40
613−20
613−12
613−10
610−80
610−40
610−20
610−12
610−10
333−80
333−40
333−20
333−12
333−10
332−80
332−40
332−20
332−12
332−10
331−80
331−40
331−20
331−12
331−10
330−80
330−40
330−20
330−12
330−10
323−80
323−40
323−20
323−12
323−10
322−80
322−40
322−20
322−12
322−10
321−80
321−40
321−20
321−12
321−10
320−80
320−40
320−20
320−12
320−10
313−80
313−40
313−20
313−12
313−10
312−80
312−40
312−20
312−12
312−10
311−80
311−40
311−20
311−12
311−10
310−80
310−40
310−20
310−12
310−10
0
5000
10000
CH4 (ppm)

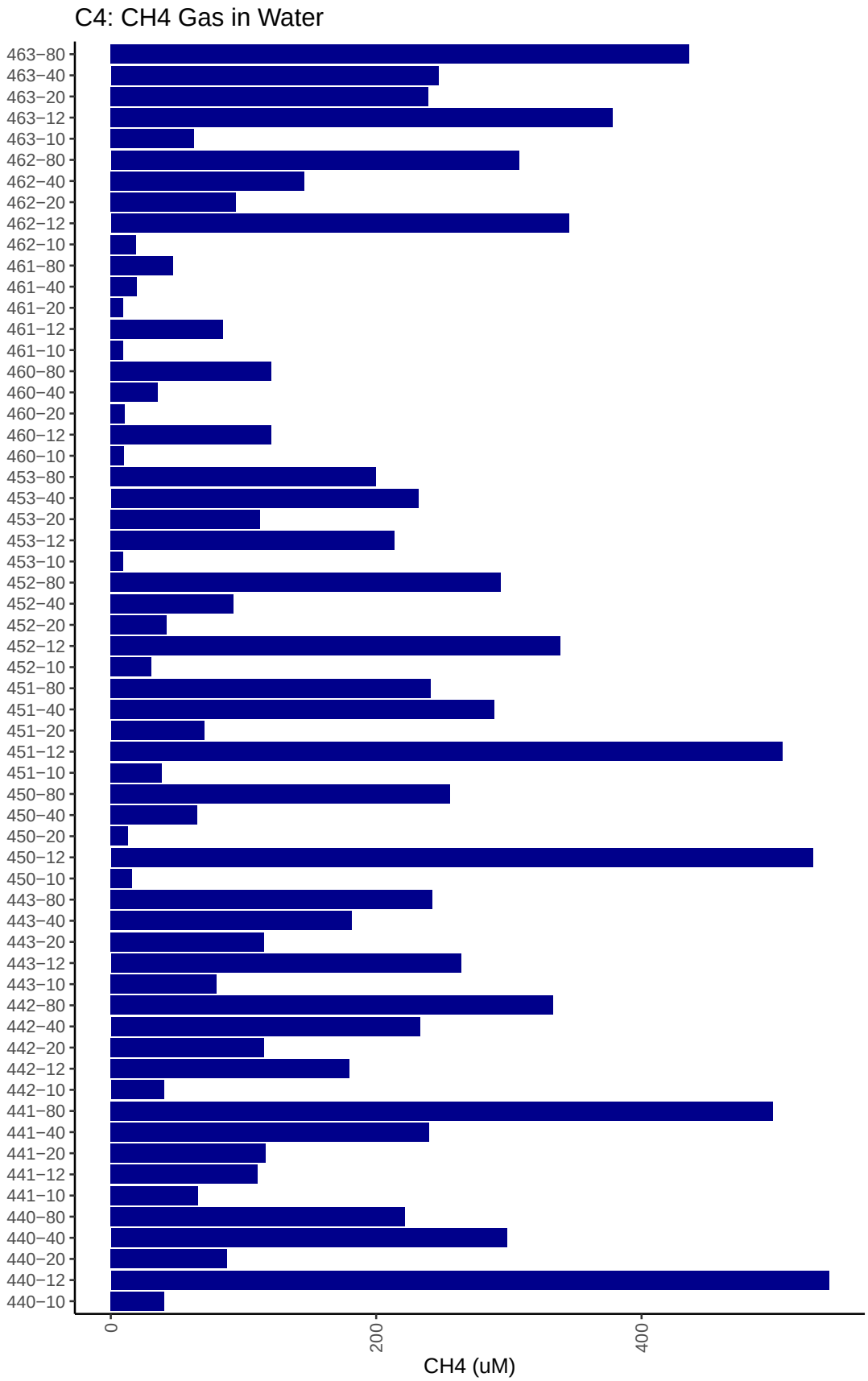
6 C4: Dilution correct samples and Check Range




CH4: Green = Within Range
463−80
463−40
463−20
463−12
463−10
462−80
462−40
462−20
462−12
462−10
461−80
461−40
461−20
461−12
461−10
460−80
460−40
460−20
460−12
460−10
453−80
453−40
453−20
453−12
453−10
452−80
452−40
452−20
452−12
452−10
451−80
451−40
451−20
451−12
451−10
450−80
450−40
450−20
450−12
450−10
443−80
443−40
443−20
443−12
443−10
442−80
442−40
442−20
442−12
442−10
441−80
441−40
441−20
441−12
441−10
440−80
440−40
440−20
440−12
440−10
0
5000
10000
CH4 (ppm)

7 Calculate gas in water




C4: CH4 Gas in Water
463−80
463−40
463−20
463−12
463−10
462−80
462−40
462−20
462−12
462−10
461−80
461−40
461−20
461−12
461−10
460−80
460−40
460−20
460−12
460−10
453−80
453−40
453−20
453−12
453−10
452−80
452−40
452−20
452−12
452−10
451−80
451−40
451−20
451−12
451−10
450−80
450−40
450−20
450−12
450−10
443−80
443−40
443−20
443−12
443−10
442−80
442−40
442−20
442−12
442−10
441−80
441−40
441−20
441−12
441−10
440−80
440−40
440−20
440−12
440−10
0
200
400
CH4 (uM)

8 Merge samples with Metadata & check all samples are present

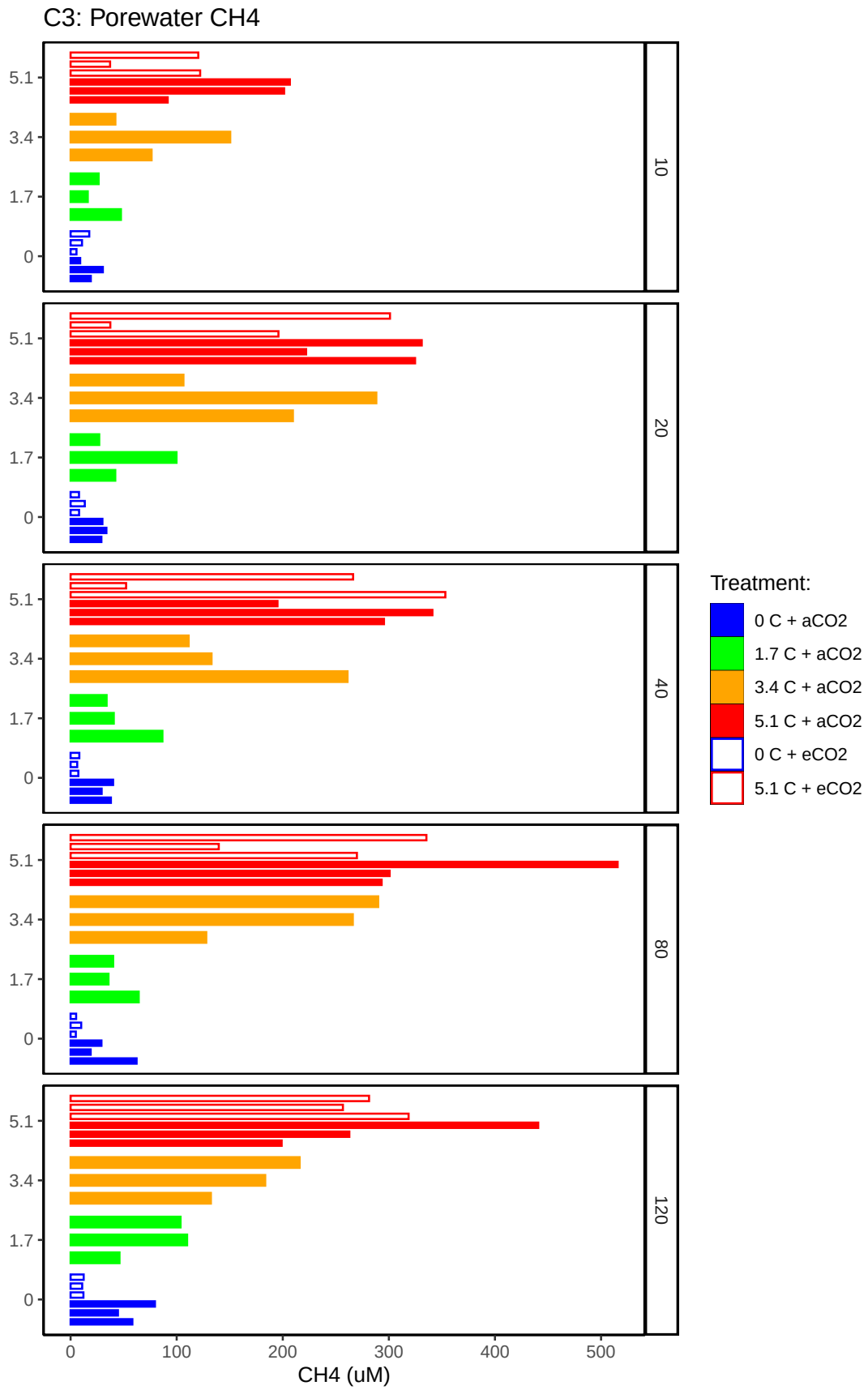
```
## Merge Samples with Metadata
```

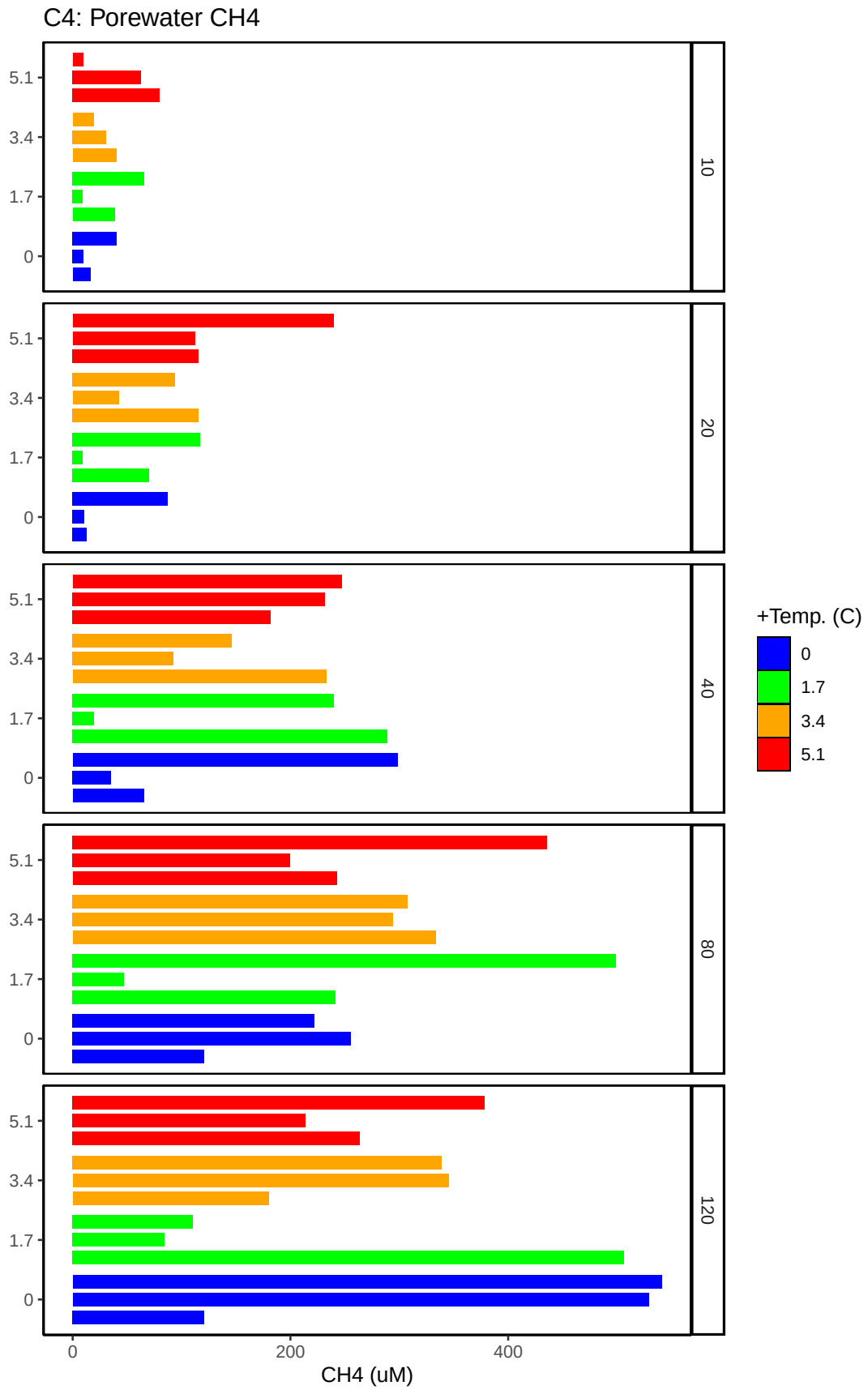
```
## All sample IDs are present in data
```

8.1 Organize Data

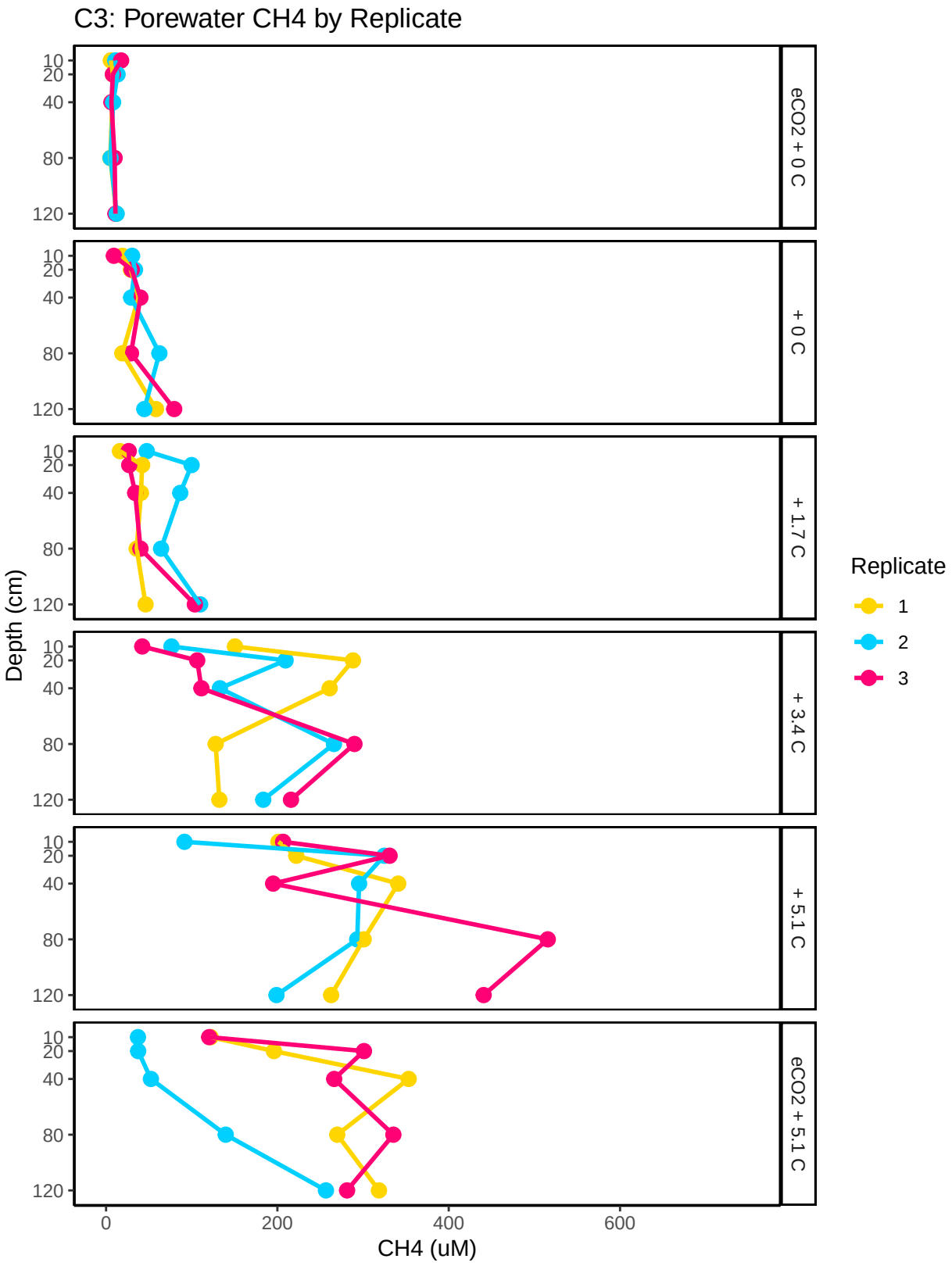
8.2 Visualize Data

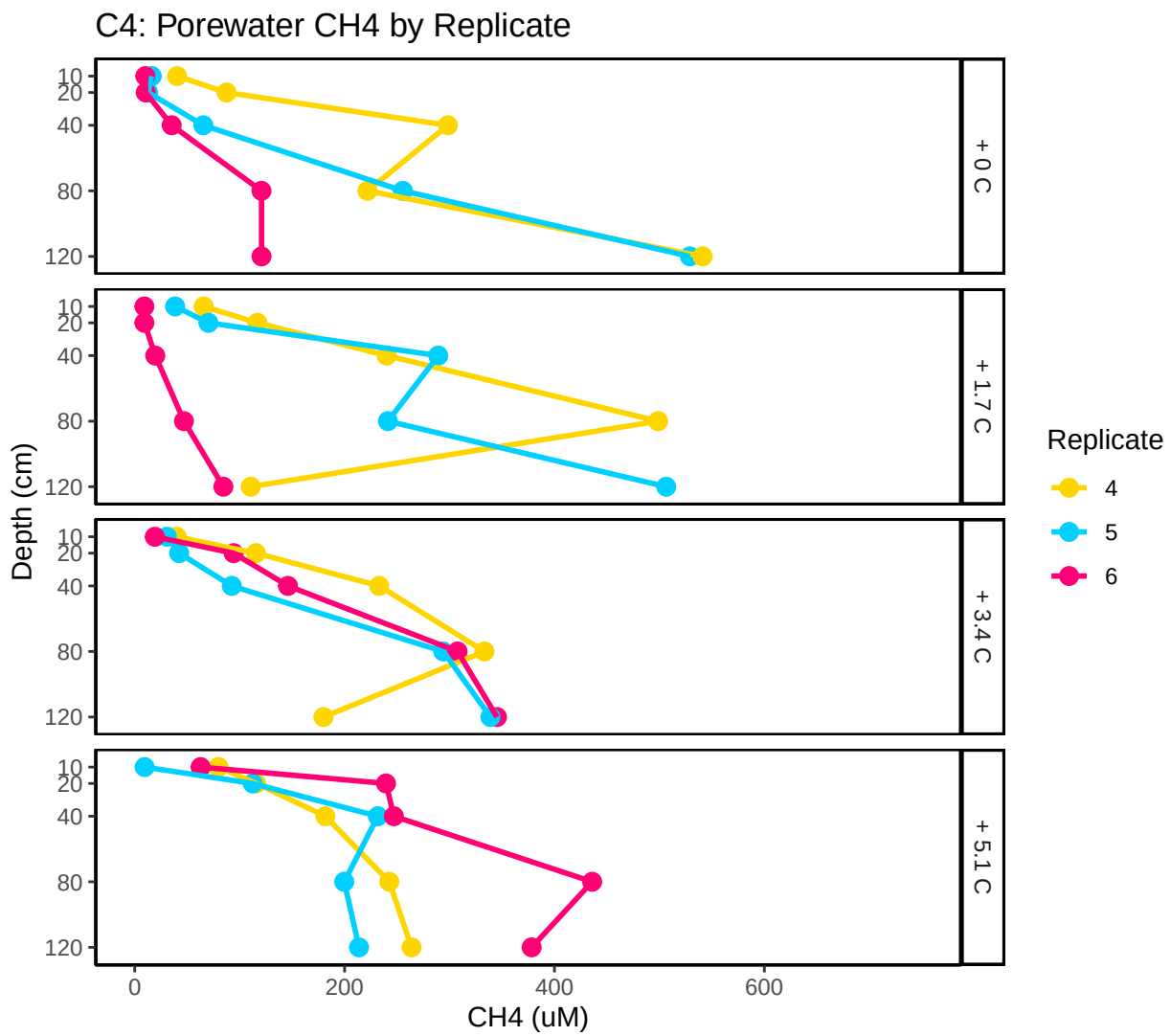
```
## Visualize Data
```


C3: Porewater CH4
5.1
3.4
1.7
0
10
20
40
80
120
0
100
200
300
400
500
CH4 (uM)
Treatment:
0 C + aCO2
1.7 C + aCO2
3.4 C + aCO2
5.1 C + aCO2
0 C + eCO2
5.1 C + eCO2


C4: Porewater CH4
5.1
3.4
1.7
0
10
20
40
80
120
0
200
400
CH4 (uM)
+Temp. (C)
0
1.7
3.4
5.1

```
## Warning: Using ‘size‘ aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use ‘linewidth‘ instead.
## This warning is displayed once every 8 hours.
## Call ‘lifecycle::last_lifecycle_warnings()‘ to see where this warning was
## generated.
```


C3: Porewater CH4 by Replicate
eCO2 + 0 C
+ 0 C
+ 1.7 C
+ 3.4 C
+ 5.1 C
eCO2 + 5.1 C
Depth (cm)
10
20
40
80
120
0
200
400
600
CH4 (uM)
Replicate
1
2
3



##Export Data

#end